

Final Answer

Microeconomic Theory
Final Exam
2023

Instructions: Answering the questions in this exam is an individual effort. The estimated answering time is provided at the beginning of each question.

1. (Approximately 55 minutes)

An agent chooses exert effort $e \geq 0$ to produce an observable output y . The output distribution $F(y, e)$ is characterized by a mean $\bar{y}(e)$ and variance $\sigma^2(e)$.

The principal proposes a contract $s(y) = a + by$ where a and b are parameters. When the observed output is y , the principal payoff is $U^P = y - s(y)$.

The expected payoff to the agent when the contract $s(y)$ is accepted, and the effort level e is

$$U^A(e, s) = a + b\bar{y}(e) - \frac{1}{2}[b^2\sigma^2(e) + e^2]$$

If the agent rejects the contract, both principal and agent receive payoff zero.

- (a) Obtain the principal's optimal choice of a and b in terms of σ^2 when output mean is given by $\bar{y}(e) = e$ and its variance $\sigma^2(e)$ is a constant, independent of the value of e .
- (b) Suppose now that the principal observes directly the agent's effort e and makes a and b depend on e . What is the optimal contract?

$$a + be - \frac{b^2\sigma^2}{2} - \frac{e^2}{2} = 0$$

$$\frac{b\sigma^2}{2} = 1 - b$$

$$b =$$

$$b \left(1 + \frac{\sigma^2}{2} \right) = 1$$

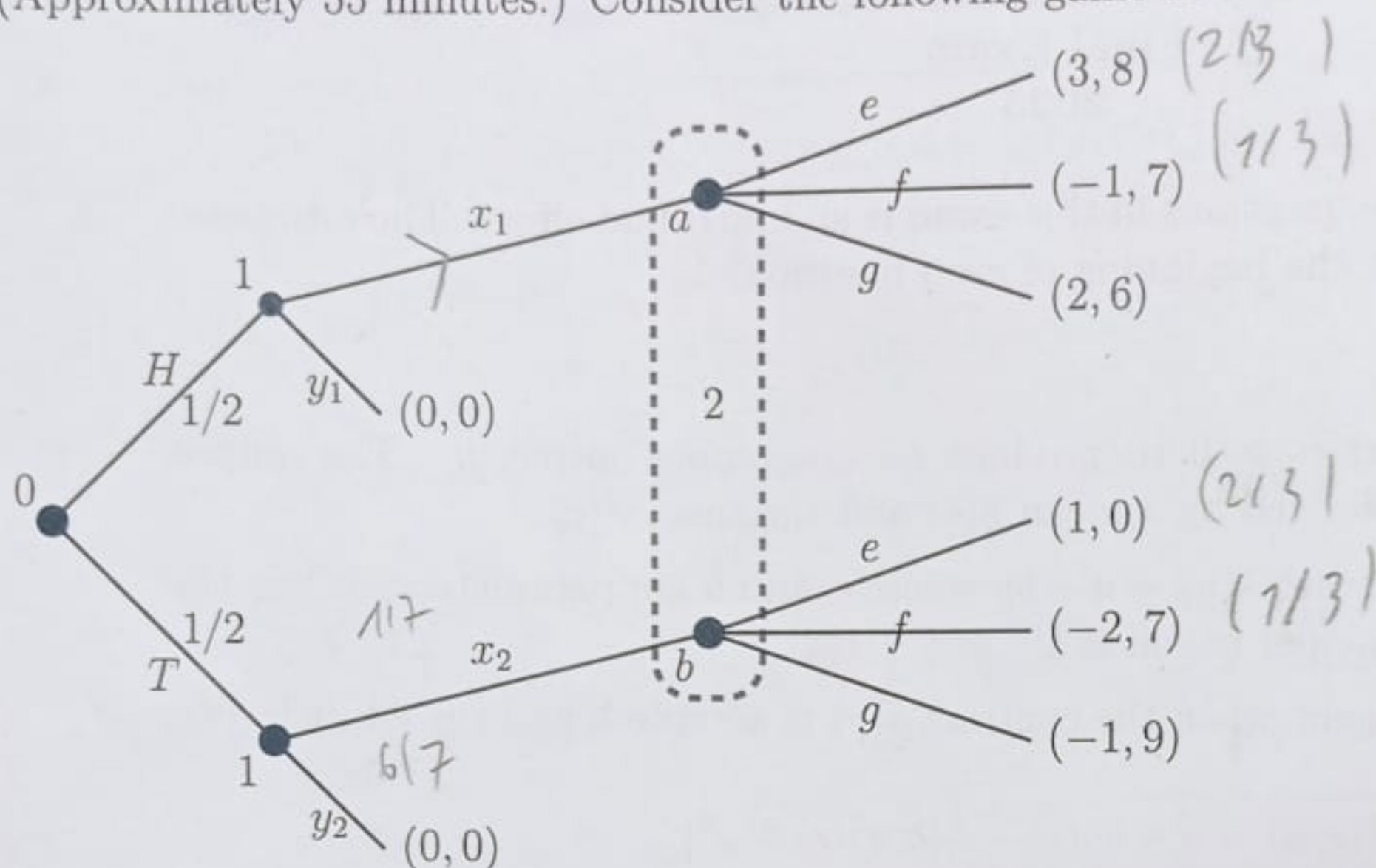
$$b \left(\frac{2 + \sigma^2}{2} \right) = 1$$

$$b = \frac{2}{\sigma^2 + 2}$$

$$b(\sigma^2 + 1) = 1$$

$$b = \frac{1}{1 + \sigma^2}$$

2. (Approximately 55 minutes.) Consider the following game in extensive form.



Nature flips a fair coin. Player 1 observes its outcome $z \in \{H, T\}$. Thus player 1 has two information sets; player 2 has a single information set, represented by a dashed box. The top node in player 2's information set is labeled a , the bottom node is labeled b . Payoffs are indicated by number pairs at terminal nodes, the first number corresponds to player 1's payoff. To fix notation, choices at each information set are labeled on the graph.

- (a) Let $\alpha \in [0, 1]$ be player 2's beliefs about being in the top node of player 2's information set. At player 2's information set, describe a function that maps every α to player 2's optimal continuation. Briefly link your answer to sequential rationality—note that one sentence suffices to establish the link.
- (b) Is the strategy profile $[(y_1, y_2), f]$ part of a sequential equilibrium? Justify your answer by providing a proof.
- (c) Is the strategy profile (σ_1, σ_2) where

$$\sigma_1(z) = \begin{cases} x_1, & \text{if } z = H \\ (1/7, 6/7) \text{ on } (x_2, y_2), & \text{if } z = T \end{cases}$$

$$\sigma_2 = (2/3, 1/3) \text{ on } (e, f)$$

part of a sequential equilibrium? Justify your answer by providing a proof.